

Project Report: UNICEF Implementing Partners Dashboard

Author: Rosella Jade A. Ricafort
Date: May 23, 2025
Environment: R Shiny
Data Source: UNICEF Transparency Portal (open.unicef.org)

1. Executive Summary

This report outlines the development and architecture of the UNICEF Implementing Partners (IP) Dashboard, an interactive decision-support system built using R Shiny. The application aggregates, normalizes, and visualizes funding distributions, regional priorities, and HACT risk ratings across global operations.

The dashboard transforms raw transparency data into interactive visual insights, helping stakeholders analyze funding concentration, compliance risks, and year-on-year financial changes.

2. Dataset Description & Normalization Engine

2.1 Raw Data Architecture

The system uses the UNICEF IP List for Publication 2025 dataset. If the dataset file is not available locally, a built-in simulation engine generates realistic data distributions (log-normal funding, regional variation, and risk categories).

Key fields:

- Partner Metadata: Partner Name, Partner Type (Government, NGO, UN Agency, Academic, Private)
 - Geographic Dimensions: Country, Region (EAPR, ECAR, ESAR, HQ, LACR, MENAR, SAR, WCAR)
 - Thematic Area: Health, Education, WASH, Child Protection, etc.
 - Financial Metrics: Cash Transfer 2023 (USD), Cash Transfer 2024 (USD)
 - Compliance Metric: Risk Rating (Low, Moderate, Significant, High)
-

2.2 Column Mapping & Data Cleaning

A flexible column-mapping function ensures the system works even if dataset column names change.

Missing values are replaced with 0.

Year-on-year growth is calculated using:

$$\text{YoY \%} = ((C_{2024} - C_{2023}) / C_{2023}) \times 100$$

3. UI/UX Design Philosophy

The dashboard uses bslib (Bootstrap 5) for a modern and structured interface.

3.1 Typography & Colors

Fonts:

- Syne → Headings and titles
- Plus Jakarta Sans → Body text and tables

Color system:

- Primary: Deep Navy (#0A2342), Royal Blue (#1976D2)
 - Risk Levels:
 - Low → Blue
 - Moderate → Light Blue
 - Significant → Amber
 - High → Crimson
-

3.2 Layout Structure

The dashboard uses `page_navbar()` with a multi-tab layout.

Filters are placed in a collapsible side panel to keep the interface clean and focused on visualizations.

4. App Features

Module 1: Overview Tab

- KPI cards (total funding, partners, countries, risk levels)
- Regional bar chart

- Donut chart for partner type distribution

Module 2: Partner Analysis

- Top 20 recipients bar chart
- Heatmap (Partner Type vs Region)
- Scatter plot (2023 vs 2024 funding comparison)

Module 3: Regional Deep Dive

- Region-based filtering
- Dynamic pie charts and summaries per region

Module 4: Risk & Compliance

- High-risk funding analysis
- Top 15 high-risk partners table

Module 5: Data Table

- Full dataset viewer
- Export options (CSV, Excel)
- Conditional formatting for risk levels

Module 6: Insights

- Automated statistical insights
- Distribution analysis and reporting

5. Analytical Models & Insights

5.1 Funding Concentration (Lorenz Curve)

Used to measure inequality in funding distribution.

$$\text{Gini} = A / (A + B)$$

Where:

- A = area between equality line and Lorenz curve
 - B = area under Lorenz curve
-

5.2 Key Insights

1. Funding is highly concentrated among top partners, showing dependency on large organizations.
 2. Data is highly skewed; a few large grants increase the average significantly.
 3. High-risk partners often receive large funding due to emergency or fragile contexts, requiring stronger monitoring systems.
-

6. How to Run the App

Step 1: Install packages

Install required R packages including shiny, bslib, plotly, DT, dplyr, tidyr, scales, shinyWidgets, readxl, and fontawesome.

Step 2: Run the dashboard

Run the application using R Shiny's runApp function with the file app.R.

If the dataset is missing, the app will automatically generate simulated data and still run normally.